1 INTRODUCTION

1.1 Project Background

1.1.1 The Hung Shui Kiu New Development Area (HSK NDA) was formerly studied under "Planning and Development Study on North West New Territories" (the NWNT Study) from 1997 to 2003. The NWNT Study was an integrated planning and engineering study to identify NDAs in North West New Territories (NWNT) in response to projections of housing demand for Hong Kong arising from the Territorial Development Strategy Review in 1996. The NWNT Study identified HSK suitable as a NDA to accommodate a population of about 160,000 and to provide 48,000 jobs upon full development. The NWNT Study identified 450 ha for the HSK NDA. Various land uses including residential, government, institution or community, education, recreation, business use, open spaces, port back-up, green belt, etc. were proposed.

1.1.2 The HSK NDA proposals were subsequently shelved in 2003 in light of an anticipated slower population growth at that time. Under the study of Hong Kong 2030: Planning Vision and Strategy (the HK2030 Study) completed by Planning Department (PlanD) in 2007, the HSK NDA identified in the NWNT Study was revisited and recommended for implementation to address the long-term housing demand and provide employment opportunities. The Chief Executive announced in his 2007-08 Policy Address the planning for the NDA in HSK as one of the ten major infrastructure projects for economic growth.

1.1.3 Having regard to the substantial changes in planning circumstances and public aspiration since completion of the NWNT Study, the boundary of the HSK NDA was tentatively expanded to 790 ha in 2011. A comprehensive planning and engineering study on the HSK NDA was commissioned in 2011.

1.1.4 The planning and engineering study is to revisit the findings and recommendations of the NWNT Study, to take into account changes in the latest circumstances and public aspiration so as to confirm the feasibility of the proposed developments to meet long-term housing, social, economic and environmental needs, and to prepare a recommended outline development plan (RODP) and preliminary engineering design for the development.

1.2 The Assignment

1.2.1 In August 2011, AECOM Asia Company Limited (AECOM) was commissioned by the Civil Engineering and Development Department (CEDD) and PlanD to undertake the HSK NDA Planning and Engineering Study (the "Study"). The Study aims to formulate a feasible land-use framework for the HSK NDA (refer to **Figure 1.1** for location plan) to meet the long-term housing and other land-use needs of Hong Kong. The Study aims to formulate an outline development plan (ODP), layout plans for the NDA, and an implementation programme. Technical assessments to confirm the feasibility and environmental acceptability of the development proposals are also required.

1.3 EIA Study Brief

Environmental Impact Assessment

Background

1.3.1 An application for an EIA Study Brief (Application No. ESB-190/2008) was lodged on 2 May 2008, based on the original 450 ha identified during the NWNT Study for the HSK NDA. An EIA Study Brief (No. ESB-190/2008) was issued on 13 June 2008. However, due to the substantial changes in planning circumstances and public aspiration since the completion of the NWNT Study in 2003, the boundary of the HSK NDA was expanded to 790 ha to allow flexibility in layout planning so as to accommodate the new demand for various land uses and associated infrastructure (**Figure 1.1**). As a result, a subsequent application for an EIA Study Brief was (Application No. ESB-221/2011) was submitted on 25 January 2011 and an EIA Study Brief (No. ESB-221/2011) was issued on 7 March 2011. The HSK NDA at that time was bounded by Tin Ying Road and Kiu Hung Road in the east;

Yuen Long Highway in the south; Lau Fau Shan Road, Deep Bay Road and hill slopes in the north; and Kong Sham Western Highway (KSWH) in the west (**Figure 1.1**).

1.3.2 Since then, the Project Proponent has been proactively conducting a series of community engagement exercises to collate views and opinions from stakeholders, green groups and local communities. In parallel to the community engagement process, the planning, design and engineering assessments of the HSK NDA have also been progressing and evolving to address various constraints and development needs as well as the comments collated in community engagement exercises. As such, the boundary of HSK NDA as presented in the previous EIA Study Brief (No. ESB-221/2011) has been refined (**Figure 1.1**) and additional associated infrastructure has also been identified to support the new town development. Details of the refinements made to the boundary of the HSK NDA and the additional infrastructure required are listed below:

i) The existing developments to the southeast of Castle Peak Road are excluded from the HSK NDA. At present, these areas are characterised by existing clustered villages, private developments and education/Government facilities. Limited land is available for further major development in this area.

ii) The existing Tin Shui Wai (TSW) Main Channel and Tin Ying Road on the east are included in the HSK NDA to regenerate the river channel and the riverside promenades. This would enhance street vibrancy and connectivity between TSW New Town and the HSK NDA.

iii) The existing KSWH is included in the HSK NDA for the construction of a new primary distributor underneath it to enhance the north-south traffic movement within the HSK NDA.

iv) The area to the west of KSWH is included in the HSK NDA for general industrial development. This would help to redistribute industrial activities to free up land at central location within the HSK NDA.

v) Fresh water and flushing water service reservoirs would be constructed at the northwestern and southeastern sides of the HSK NDA with proposed reuse of reclaimed water.

1.3.3 As a result of the above changes, the total area of the HSK NDA has been reduced from about 790 ha to 714 ha. As the revision to the boundary of the HSK NDA might be considered a key change that in turn might alter the previous scope of the Project, a revised Project Profile (No. PP-531/215) was submitted to the Environmental Protection Department (EPD) on 30 November 2015 to apply for a new EIA Study Brief. A new EIA Study Brief (No. ESB–291/2015) for the Project was issued under the EIAO in January 2016.

1.3.4 In accordance with Section 5.1 of the EIA Study Brief, a summary highlighting the relevant EIA sections fulfilling the respective requirements of the EIA Study Brief and EIAO-TM (in particular Annexes 11 and 20 of EIAO-TM) is presented in **Appendix 1.1**.

1.4 Revised Recommended Outline Development Plan

1.4.1 The overall vision of the HSK NDA (herein referred to as the “Project”) is to build a sustainable, people-oriented and balanced living and working community for Hong Kong people. The Project will not only be a localised residential and employment cluster, but also a regional centre serving as the commercial and civic hub of the NWNT. In the territorial context, the Project will give impetus to foster Hong Kong’s economic growth.

1.4.2 The Revised RODP (presented in **Figure 2.2** and described in **Section 2.4**) has been developed to include the following key features:

- The major town centre featuring mega commercial complexes of offices, hotels and shopping malls, mixed commercial and high density residential developments, a civic node with Government offices and magistrates' court, and a Regional Plaza is planned around the proposed Hung Shui Kiu Station along existing West Rail Line (HSK Station) to form the 'Regional Economic and Civic Hub' in the NWNT.

- A secondary node with offices, hotels, shopping malls and high density residential developments is positioned near the existing TSW Station to act as a district commercial centre and a catalyst for the revitalisation of TSW New Town.

- To optimise land for residential purposes to meet acute housing needs of the Hong Kong people, areas skirting the commercial nodes and areas in the north and northeastern parts of the Project area are proposed for public and private residential developments with various social and community facilities which would integrate well with the adjoining TSW New Town and Lau Fau Shan area.

- By concentrating commercial and residential developments around the proposed HSK Station and existing TSW Station, and through the introduction of Environmentally Friendly Transport Services (EFTS) to provide the feeder services, most of the population and employment will be within walking distance of a public transport node.

- Land has been reserved for construction of public housing for local rehousing purposes to help maintain the social fabric of the existing communities.

- A "Logistics, Enterprise and Technology Quarter" to accommodate a logistics park and an "Enterprise and Technology Park" ("E+TP"), an Industrial Zone, as well as land for "Port Back-up, Storage and Workshop Uses" ("PBU+SWU") is planned in the northwestern part of the Project near the KSWH to grasp the locational advantages of being linked by existing and planned highways to the Hong Kong International Airport, different parts of Hong Kong and Shenzhen. It is envisaged to develop the Project to be a strategic employment hub which offers a wide range of employment opportunities. This important economic and employment node will be efficiently linked up with the West Rail stations, the commercial nodes and residential clusters by a comprehensively planned EFTS.

- Adoption of development intensity and a stepped building height profile where the tallest and densest developments are concentrated at the commercial nodes and descending towards the Lau Fau Shan and Deep Bay area. Development intensity and building height will be gradually reduced in the major town centre, to the secondary node at the residential area in the Lau Fau Shan area.

- Based on the unique, local demographic and social characteristics, sites in suitable locations are reserved for various "Government, Institution or Community" ("G/IC") uses such as magistrates' court, hospital, sports ground, sports complexes and other social welfare and recreational facilities. It is aimed to create a pleasant liveable residential environment serviced by readily accessible social, community and recreational facilities. Particular attention should be paid to ensure timely provision of various social and community facilities and infrastructure in tandem with the population intake of the Project.

- To promote walkability and street vibrancy, three shopping streets lined with retail frontages are proposed along the western bank of TSW Main Channel, within the residential area linking the major town centre and "E+TP", and along the pedestrian street linking the civic node with the proposed HSK Station.

- Existing traditional village clusters within the Project area will remain while allowing future growth in accordance with the current statutory plans.

- Where possible, all natural and landscape features are integrated into the layout of the Project development. For example, the planning framework envisages the retention of the San Sang San Tsuen egretry and its flight path by designating these areas as "Green Belt" ("GB") and "Local Open Space" ("LO"), respectively. Important elements of cultural and historical resources will also be preserved in the Project area.

- A Regional Town Park together with major recreational facilities is located in the centre of the Project to create a social and recreational hub. An integrated open space, pedestrian walkway and cycle track network is proposed mainly along the river channel to form a green spine and major breezeway penetrating the Project area in a north-south direction. It also enables better connection with the neighbouring TSW New Town.

- Two Fung Shui lanes are maintained which provide two intersecting east-west and northeast-southwest visual corridors connecting to the Ping Shan heritage precinct. All cultural heritage structures are retained and a cultural heritage trail is envisaged to promote the presence of these features.

- In addition to the West Rail Line, Light Rail Transit and the road-based transport system, mobility and connectivity within the Project area and with nearby new towns are also enhanced by the EFTS, an artery cycleway, a comprehensive pedestrian walkway and cycle track network.

- A number of smart and green city initiatives will be planned and explored to make the Project a smart and green place to live, work, learn and play.

1.5 Designated Projects

1.5.1 The Project is a designated project (DP) under Item 1 Schedule 3 of Environmental Impact Assessment Ordinance (EIAO) - Engineering feasibility study of urban development projects with a study area covering more than 20 ha or involving a total population of more than 100,000, as it covers an area of 714 ha and will accommodate a total population of 218,000.

1.5.2 In addition, based on the Revised RODP, the Project would comprise the following DPs by virtue of items A.1, A.2, A.3, A.8, A.9, B.5, F.1, F.3(b), F.4, G.2 and Q.1 of Schedule 2 of the EIAO (**Table 1.1** and **Figure 1.2** and **Figure 1.3**). A description of the DPs is given in **Section 2.4**.

Table 1.1 Schedule 2 Designated Projects in the HSK NDA

Designated Project Reference No.	Schedule 2 Designated Project		Work Component / Reference in Revised RODP
DP1 [1]	**Part I, A.1**	**A road which is an expressway, trunk road, primary distributor road or district distributor road including new roads, and major extensions or improvements to existing road**	**Construction of new primary distributor road (Road P1)**
DP2 [1]	**Part I, A.1**	**A road which is an expressway, trunk road, primary distributor road or district distributor road including new roads, and major extensions or improvements to existing road**	**Construction of eight new distributor roads (Road D1 to D8)**
DP3 [2]	Part I, A.2	A railway and its associated stations	Construction of new West Rail HSK Station (Site 4-34)
DP4 [2] (Potential DP)	Part I, A.3	A tramway and its associated stations	Construction of EFTS – subject to further review
DP5 [1]	**Part I, A.8**	**A road or railway bridge more than 100 m in length between abutments**	**Construction of slip roads between: Road D8 Junction and existing Castle Peak Road; Junction of D8/P1 and Junction of D7/P1; and KSWH connection to Road D3**
DP6 [1]	**Part I, A.9**	**A road fully enclosed by decking above and by structure on the sides for more than 100 m**	**Construction of partly depressed and partly decked-over roads located at Road D2, Road D4, and Road D6**
DP7 [2] (Potential DP)	Part I, B.5	A container back-up area, container storage, container handling or container packing area (including a container vehicle parking area) more than 5 ha in size and within 300 m of an existing or planned receiver	Construction of a new container back-up and storage area (Sites 3-1, 3-4, 3-5, 3-13 and 3-14) – subject to further review
DP8 [2]	Part I, F.1	Sewage treatment works with an installed capacity of more than 15,000 m^3 per day	Construction of new HSK Sewage Treatment Works (STW) (Site 3-26 and part of existing San Wai STW)
DP9 [1]	**Part I, F.3(b)**	**A sewage pumping station –** **(b) with an installed capacity of more than 2,000 m^3 per day and a boundary of which is less than 150 m from an existing or planned receiver**	**Construction of four new sewage pumping stations (Sites 2-34, 3-41, 3-48 and 4-35)**
DP10 [2]	Part I, F.4	An activity for the reuse of treated sewage effluent from a treatment plant	Construction of flushing water service reservoirs for reuse of reclaimed water at Tan Kwai Tsuen and Fung Kong Tsuen (Sites 5-40 and 3-3)

Designated Project Reference No.	Schedule 2 Designated Project		Work Component / Reference in Revised RODP
DP11 [2]	Part I, G.2	A refuse transfer station	Construction of one refuse transfer station (Site 3-12)
DP12 [1]	**Part 1, Q.1**	**All projects including new access roads, railways, sewers, sewage treatment facilities, earthworks, dredging works and other building works partly or wholly in an existing or gazetted proposed country park or special area, a conservation area, an existing or gazetted proposed marine park or marine reserve, a site of cultural heritage, and a site of special scientific interest.**	**Construction of Road P1 and slip-road from KSWH to Road D3 partly located within the "Conservation Area" of Yuen Tau Shan**

Note:
1 Subject to an Environmental Permit application under this EIA Study.
2 Subject to separate EIA Study, as required.

Non-Designated Projects

1.5.3 Apart from the Scheduled 2 DPs (**Table 1.1**), there are also a number of non-DP elements in the Revised RODP including but not limited to the following sub-elements summarised in **Table 1.2**. Details of these non-DP elements are presented in **Section 2.4** and their locations are shown in the Revised RODP (**Figure 2.2).**

Table 1.2 Summary of Non-Designated Projects

Non-Designated Project	Sub Element
Residential Development	Special Residential – Public Rental Housing (with commercial) (RSc)
	Special Residential – Public Rental Housing (with Commercial) (RSc*)
	Subsidised Sales Flat (with Commercial) (R2(SSF)c
	Subsidised Sales Flat (with Commercial) (R2(SSF)c*
	Residential Zone 1 (with Commercial) (R1c)
	Residential Zone 2 (R2)
	Residential Zone 2 (with Commercial) (R2c)
	Residential Zone 2 (R2*)
	Residential Zone 3 (R3)
	Residential Zone 3 (with Commercial) (R3c)
	Residential Zone 3 (R3*)
	Rural Residential Zone 4 (RR4)
	Residential Zone 4 (R4*)
Government	-
Institution or Community	-
Education	Primary and Secondary Schools
Open Space	Regional Open Space
	District Open Space
	Local Open Space
Amenity	-

Non-Designated Project	Sub Element
Other Specified Uses	Commercial and Residential
	Enterprise and Technology Park
	Logistics Facility
	Port Back-Up, Storage and Workshop Uses (under 5 ha in area and not within 300 m of an existing or planned receiver)
	Regional Plaza
	Education and Related Facilities
	Public Transport Interchange
	Refuse Collection Point, Electricity Substation, Petrol Filling Station, Liquefied Petroleum Gas Filling Station, Flood Retention Facilities, Emergency Access Point
Industry	-
Roads	Local roads

* Existing/planned/committed developments

1.5.4 If there are any future change(s) after the approval of this EIA Report, the following steps will be taken:

i) Should the change(s) involve a DP item under Schedule 2 of the EIAO, the requirements under the EIAO will be complied with; and

ii) Should the change(s) not involve any DP items under Schedule 2 of the EIAO, prevailing planning mechanisms and standards will be followed and relevant EIA findings will be conformed to.

Objectives of the EIA Report

1.5.5 The purpose of the EIA study is to provide information on the nature and extent of environmental impacts arising from the construction and operation of the Project and associated works that take place concurrently, the constraints imposed by the existing/planned/committed developments in the Project area and land use zonings shown on the relevant Outline Zoning Plans (OZP), including the latest Ha Tsuen OZP, Ping Shan OZP, Lau Fau Shan & Tsim Bei Tsui OZP, TSW OZP, Tong Yan San Tsuen OZP, and Lam Tei and Yick Yuen OZP.

1.5.6 The objectives of the EIA study are as follows:

i) to describe the Project and associated works together with the requirements and environmental benefits for carrying out the Project and associated works;

ii) to identify and describe the elements of the community and environment likely to be affected by the Project and associated works and/or likely to cause adverse impacts to the Project, including both the natural and man-made environment and the associated environmental constraints;

iii) to provide information on the consideration of alternatives to avoid or minimise the potential adverse environmental impacts on sensitive receivers and sensitive uses in the Project site and adjacent areas that may be subject to: (i) the adverse environmental impacts of the Project and associated works, and/or (ii) the adverse impacts of the existing/committed/planned developments in the Project site and adjacent areas, in particular the KSWH, Yuen Long Highway, West Rail and Light Rail projects including associated committed/planned infrastructure provisions (such as HSK Station) and improvement/expansion schemes, if any; the proposed Tuen Mun Western Bypass; and developments proposed under “The Study on Enhancement of

the Lau Fau Shan Rural Township and Surrounding Areas"; to compare the environmental benefits and dis-benefits of each of different options; to provide justifications and set out constraints for selecting the preferred option(s) and to describe the part that environmental factors played in the selection;

iv) to identify and assess air quality impact, noise impact, water quality impact, waste management implications, sewerage impact assessment, ecological impact, fisheries impact, impact on cultural heritage, and landscape and visual impact; quantify emission sources and determine the significance of impacts on sensitive receivers and potential affected uses;

v) to identify and quantify any potential losses or damage to flora, fauna, natural habitats and to propose measures to mitigate these impacts as necessary;

vi) to provide tree and vegetation survey information in accordance with the EIAO Guidance Note No. 6/2010. The tree treatment proposal shall include plans showing the locations of tress retained/transplanted/felled and recipient locations for the trees identified for transplant or compensation;

vii) to identify any negative impacts on cultural heritage and to propose measures to mitigate these impacts;

viii) to identify the negative impacts and propose measures for avoidance or the provision of infrastructure or mitigation measures to minimise pollution, environmental disturbance and nuisance during construction and operation of the Project and associated works;

ix) to investigate the feasibility, practicability, effectiveness and implications of the proposed impact avoidance and/or mitigation measures;

x) to identify, predict and evaluate the residual (i.e. after avoidance or practicable mitigation) environmental impacts and the cumulative effects expected to arise during the construction and operation phases of the Project and associated works in relation to the sensitive receivers and potential affected uses;

xi) to identify, assess and specify methods, measures and standards to be included in the detailed design, construction and operation of the Project and associated works which are necessary to mitigate these residual environmental impacts and cumulative effects and reduce them to the acceptable levels;

xii) to investigate the extent of the secondary environmental impacts that may arise from the proposed mitigation measures and to identify constraints associated with the mitigation measures recommended in the EIA study, as well as provision of any necessary modification;

xiii) to identify individual project(s) proposed under the Study which constitutes DPs under Schedule 2 of the EIA Ordinance; to specify the proposed Schedule 2 DP(s) that would apply for environmental permits with reference to this EIA Report and DP(s) that would be subject to further EIA study; and for the latter to identify the outstanding issues that need to be addressed in the EIA study;

xiv) to design and specify the environmental monitoring and audit requirements; and

xv) to identify any additional studies necessary to implement the mitigation measures of monitoring and proposals recommended in the EIA Report.

1.6 Structure of the EIA Report

1.6.1 This document presents the findings of the EIA under the following sections:

Chapter	Title	Aims
1	Introduction	Introduces the project background and objectives of this Report
2	Project Description and Consideration of Alternatives	Provides a description of the Project (Revised RODP) and summarises the consideration and selection of alternative options for development layout and construction methodologies
3	Air Quality Impact	Presents the legislation, baseline conditions, assessment methodology, impact assessment and mitigation measures required for air quality impacts
4	Noise Impact	Presents the legislation, baseline conditions, assessment methodology, impact assessment and mitigation measures required for noise impacts
5	Water Quality Impact	Presents the legislation, baseline conditions, assessment methodology, impact assessment and mitigation measures required for water quality impact
6	Sewerage and Sewage Treatment Implications	Presents the legislation, baseline conditions, assessment methodology, impact assessment and mitigation measures required for sewerage and sewage treatment
7	Waste Management Implications	Presents the legislation, baseline conditions, assessment methodology, impact assessment and mitigation measures required for waste management
8	Land Contamination Impact	Presents the legislation, baseline conditions, assessment methodology, impact assessment and mitigation measures required for land contamination evaluation
9	Ecological Impact	Presents the legislation, baseline conditions, assessment methodology, impact assessment and mitigation measures required for ecological impacts
10	Fisheries Impact	Presents the legislation, baseline conditions, assessment methodology, impact assessment and mitigation measures required for fisheries impacts
11	Landscape and Visual Impact	Presents the legislation, baseline conditions, assessment methodology, impact assessment and mitigation measures required for landscape and visual impacts
12	Impact on Cultural Heritage	Presents the legislation, baseline conditions, assessment methodology, impact assessment and mitigation measures required for cultural heritage impacts

Chapter	Title	Aims
13	Environmental Monitoring and Audit Requirements	Summarises the EM&A requirements
14	Summary of Environmental Outcomes	Summarises the key environmental outcomes arising from the EIA study
15	Conclusion	Summarises the environmental impacts identified in the EIA